

MODULE 2: CPM & PERT

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CPM (Critical Path Method)

It is a deterministic project management technique that focuses on identifying the critical path, which is the longest sequence of tasks in a project that must be completed on time to avoid delaying the entire project.

1. Used for projects with well-defined and predictable activities.
2. It is used when task durations are known and fixed.
3. Focuses on the critical path, the longest sequence of dependent activities.
4. Task durations are fixed or deterministic.
5. Emphasizes time-cost trade-offs, allowing for crashing of activities (speeding up tasks to reduce project duration).
6. Suitable for projects where time and cost are crucial.

Applications:

1. Construction projects (buildings, roads).
2. Industrial production or manufacturing processes.
3. Projects with clear time constraints and cost targets.
4. Software development projects with known requirements.

Advantages of CPM:

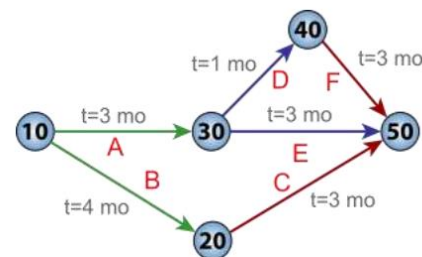
1. **Time and Cost Management:** CPM allows managers to balance time and cost, making it ideal for projects where both are critical. Activities on the critical path can be accelerated by allocating more resources (crashing).
2. **Resource Allocation:** It helps allocate resources effectively to critical tasks, ensuring that project deadlines are met.
3. **Simpler for Deterministic Projects:** CPM is suited for projects with fixed, predictable timelines, making it easier to plan and control.
4. **Clear Visualization:** The method provides a clear picture of dependencies, task durations, and critical/non-critical activities.
5. **Focus on Critical Path:** Identifies the most important tasks (on the critical path), allowing project managers to prioritize these tasks and avoid delays.

Disadvantages of CPM:

1. **Limited for Uncertain Projects:** CPM is less effective in projects with uncertain or variable task durations, where time estimates cannot be accurately predicted.
2. **Requires Accurate Time Estimates:** Like PERT, CPM heavily relies on accurate time estimates, and any errors in input can affect the entire project plan.
3. **Neglects Uncertainty:** Since CPM is deterministic, it does not handle uncertainty well, which can lead to issues if unexpected delays occur.
4. **Can Be Overwhelming:** For large and complex projects with numerous tasks, the CPM diagram can become overly complicated and difficult to manage.

PERT (Program Evaluation and Review Technique)

PERT is a probabilistic project management technique that uses a statistical approach to estimate task durations, particularly in uncertain or research and development projects. It considers optimistic, pessimistic, and most likely time estimates for each task, providing a range of possible project durations.



A PERT network chart for a seven-month project with five [milestones](#) (Nodes: 10 through 50) and six activities (Arrows: A through F)

1. Used for projects with uncertain or unpredictable activities.
2. Relies on probabilistic time estimates (optimistic, most likely, and pessimistic times).
3. Suitable for research and development, new product launches, or innovative projects.
4. Focuses on managing uncertainties in project scheduling.
5. Helps in calculating the probability of completing the project within a given time frame.

Applications:

1. Research and development projects.
2. New product development.
3. Projects where task durations are uncertain or difficult to estimate.
4. Space and defence projects.

Advantages of PERT:

1. **Handles Uncertainty:** PERT is ideal for projects with uncertain time estimates, as it allows for probabilistic modelling using optimistic, pessimistic, and most likely time estimates.
2. **Improves Planning:** It helps managers plan complex projects by breaking them into smaller tasks and understanding dependencies.
3. **Better Risk Management:** By using time estimates with probabilities, PERT helps in assessing the risk of delays and uncertainties.
4. **Flexible Scheduling:** It allows for flexibility in scheduling by recalculating timelines based on changing conditions.
5. **Identifies Critical Tasks:** PERT highlights the critical path and activities that need close attention, helping avoid bottlenecks.

Disadvantages of PERT:

1. **Complex and Time-Consuming:** Setting up and maintaining a PERT chart for large projects with many activities can be complicated and time-consuming.

2. **Inaccurate Time Estimates:** Since PERT relies heavily on time estimates, inaccurate input can lead to faulty project planning and predictions.
3. **Focus on Time, Not Cost:** PERT primarily emphasizes time estimation and management, without much focus on cost analysis or optimization.
4. **Limited Usefulness for Simple Projects:** For projects with well-defined tasks and certain timelines, PERT may be overcomplicated and unnecessary.

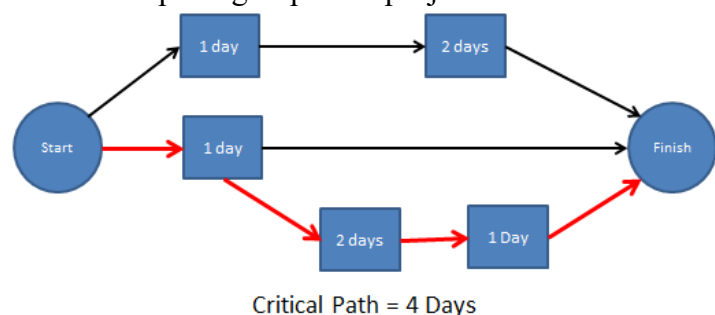
Steps in PERT

1. **Identify Activities and Milestones:** List tasks (e.g., "design," "testing") and milestones (e.g., "design approved") for the project.
2. **Determine Sequence:** Order tasks logically, ensuring dependencies are clear (e.g., "testing" follows "prototype").
3. **Create Network Diagram:** Build a flowchart showing task sequence and dependencies with nodes and arrows.
4. **Estimate Time:** Use optimistic (O), pessimistic (P), and most likely (M) estimates to calculate expected time with the PERT formula.
5. **Find Critical Path:** Identify the longest path in the diagram that dictates project completion time.
6. **Update PERT Chart:** Regularly revise the chart to reflect progress, adjust estimates, and manage delays.

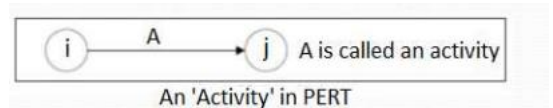
Aspect	CPM	PERT
Nature	Deterministic (fixed time estimates)	Probabilistic (variable time estimates)
Focus	Time-cost trade-offs	Time uncertainty
Application	Repetitive and construction projects	Research and development projects
Task Duration	Known and predictable	Uncertain and estimated
Activity Types	Focuses on critical and non-critical activities	Focuses on probabilistic paths
Management Emphasis	Time and cost optimization	Time management and risk analysis
Path Identification	Identifies a single critical path	May involve multiple potential paths

Key terms in Network Diagram

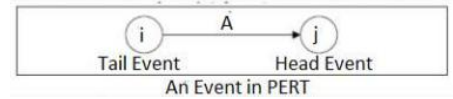
1. **Network Diagram:** it is a graphical representation of logical sequence in which different activities are interrelated to each other while completing a specific project. The numerical value(nodes) here is event and the edges here are activities.



2. **Activity:** An activity is a task or work element that consumes time and resources in a project. In the network diagram, activities are represented as arrows (in Activity-on-Arrow) or nodes (in Activity-on-Node). **Example:** Developing a prototype, preparing a report, or conducting a test.

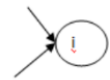


3. **Event:** An event signifies the beginning or end of an activity in the project. Events do not consume time or resources; they are milestones. In Activity-on-Arrow diagrams, events are represented as circles or nodes at the start or end of an activity. **Example:** Completion of a design phase or approval of a report.

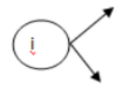


4. **Predecessor Activity:** An activity that must be completed before another activity can start. In network diagrams, a predecessor activity is the one that comes before a dependent activity and is linked with arrows. **Example:** Completing the foundation before starting the construction of walls.
5. **Successor Activity:** An activity that cannot start until a predecessor activity is completed.: It follows a predecessor activity and relates to arrows. **Example:** Painting the walls after completing construction.
6. **Node:** A node is a point in the network diagram representing an activity (in Activity-on-Node) or an event (in Activity-on-Arrow). It is depicted as a box or circle and shows the start or end of activities. **Example:** In an Activity-on-Node diagram, a node represents a specific activity like "Design Complete."
7. **Critical Path:** The longest sequence of activities in the network that determines the minimum project duration. If any activity on this path is delayed, the entire project is delayed. The critical path is highlighted in the network diagram and is usually marked to show it is critical. **Example:** A series of dependent tasks such as "Design → Build → Test" might form a critical path.
8. **Earliest Start Time (EST):** The earliest time an activity can begin, based on the completion of all its predecessor activities. Calculated during a forward pass through the network diagram. **Example:** If an activity has an EST of day 5, it cannot start before the 5th day of the project.
9. **Earliest Finish Time (EFT):** The earliest time an activity can be completed, calculated by adding the activity's duration to its earliest start time. It is determined during the forward pass. **Example:** If an activity takes 4 days and has an EST of day 5, its EFT will be day 9.
10. **Latest Start Time (LST):** The latest time an activity can start without delaying the project's completion. Calculated during the backward pass through the network diagram. **Example:** If the LST of an activity is day 8, it must begin no later than this to avoid project delays.
11. **Latest Finish Time (LFT):** The latest time an activity can be completed without delaying the project's completion time. It is determined during the backward pass. **Example:** If an activity has an LFT of day 12, it must finish by that day.

12. Merge event: An event in which more than one activity comes and join is called as merge event. The event is set to occur only when each activity joining it is completed.



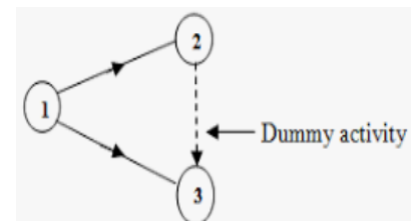
13. Burst event: an event from which more than one activity originated is known as burst event. Each activity then originates from Burst event can be started only when burst event is reached or when bursts event has occurred.



14. Forward Pass: A method used in network diagrams to calculate the earliest start and finish times for each activity. It moves from the start to the end of the project. Helps in determining the project's earliest completion time.

15. Backward Pass: A method used to calculate the latest start and finish times for each activity. It moves from the project's end to the start. Helps in identifying slack and the critical path.

16. Dummy Activity: A dummy activity is used to show logical relationships between tasks when no actual work is being done. It represents dependency but does not consume time or resources. Depicted as a dashed line in Activity-on-Arrow diagrams. **Example:** If task A must be completed before tasks B and C, but there is no real work between A and B, a dummy activity may be used to show this dependency.



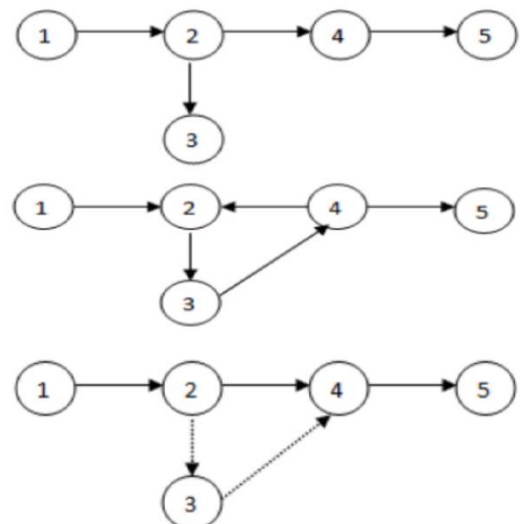
17. Optimistic Time (O), Pessimistic Time (P), and Most Likely Time (M) (PERT-specific terms): These time estimates are used in PERT to calculate expected task durations under uncertainty. Used to calculate the Expected Time (ET) of activities in PERT using the formula:

$$ET = (O + 4M + P) / 6$$

- **Optimistic Time (O):** The shortest possible time in which the activity can be completed.
- **Pessimistic Time (P):** The longest possible time the activity could take.
- **Most Likely Time (M):** The best estimate of the activity duration based on normal circumstances.

18. Common errors in drawing Network

1. **Dangling:** If a certain activity gets disconnected before the completion of all the activities in the network, then dangling error occurs. An error of dangling goes against the rule that there should be only one end even that is event from which no activity originates.
2. **Looping:** Sometimes due to faulty network sequence, network cannot proceed i.e. endless loop is created in a network. Looping error is seen in events 2, 3, and 4. This situation can be avoided by checking the precedence relationship of all the activities and by numbering them in logical sequence.



3. **Redundancy:** Presence of unnecessary dummy activity in a network logic is known as redundancy.

Float

Float (also known as slack) refers to the amount of time an activity in a project can be delayed without affecting the overall project timeline or the completion of subsequent activities. Floats are an important concept in project management, particularly in **CPM (Critical Path Method)**, as they help identify the flexibility in scheduling non-critical activities.

Types of Floats:

1. **Total Float:** The total amount of time that an activity can be delayed without delaying the overall project completion. If the total float is zero, the activity is on the **critical path**, meaning it cannot be delayed without delaying the project. **Formula:**

$$\text{Total Float} = \text{LFT} - \text{EFT or LST} -$$

2. **Free Float:** The amount of time an activity can be delayed without delaying the start of the next (immediate successor) activity. Free float focuses on the relationship between two directly linked tasks. Unlike total float, it does not impact subsequent activities beyond the immediate successor. **Formula:**

$$\text{Free Float} = \text{EST of successor} - \text{EFT of current activity}$$

3. **Independent Float:** The time an activity can be delayed without affecting the start of any other subsequent activities, assuming all predecessor activities finish at their latest times. This float provides complete freedom for the activity without relying on any other activities. **Formula:**

$$\text{Independent Float} = \text{EST of successor} - \text{LFT of Predecessor} -$$

4. **Interfering Float:** The portion of total float that interferes with subsequent activities. It is the difference between total float and free float. This float, if used, will delay the next activity but not the project. **Formula:**

$$\text{Interfering Float} = \text{Total Float} - \text{Free Float}$$

Latest Finish Time (LFT)
Earliest Finish Time (EFT)
Latest Start Time (LST)
Earliest Start Time (EST)
Activity Duration (AD)

Determination of Floats:

The process of determining float involves calculating the earliest and latest start/finish times for each activity. This is done using **forward and backward passes** in the network diagram:

1. **Forward Pass:** Calculates the **Earliest Start Time (EST)** and **Earliest Finish Time (EFT)** of each activity, moving from the start to the end of the project.
2. **Backward Pass:** Determines the **Latest Start Time (LST)** and **Latest Finish Time (LFT)** for each activity, moving from the end of the project to the start.

Calculation Example: EST = 5 days, EFT = 10 days, LST = 7 days, LFT = 12 days, AD=4 days, EFT of Successor = 13 days

- Total Float=LST-EST=7-5=2 days

- Free Float=EST of successor–EFT of current=13–10=3 days
- Independent Float = EST of successor – LFT of predecessor – Activity Duration = 13–5–4=4 days

Crashing of Network

Crashing in project management refers to a technique where the project's timeline is shortened by reducing the duration of critical activities. This is typically done by allocating additional resources (such as manpower, equipment, or overtime) to speed up completion. Crashing is used when a project is behind schedule or needs to meet a deadline. **Key Points:**

1. **Applies to Critical Path:** Crashing focuses only on activities on the critical path (activities with zero total float), as reducing the duration of non-critical activities will not shorten the overall project time.
2. **Cost-Time Trade-Off:** Crashing increases the cost of the project since extra resources are utilized. Therefore, managers aim to achieve the maximum time reduction with minimal cost increases.
3. **Crashing is Incremental:** Activities are crashed one at a time, starting with the least expensive option, until the desired project duration is reached or no further crashing is cost-effective.

Steps to Perform Crashing:

1. **Identify Critical Activities:** List the tasks on the critical path.
2. **Calculate the Cost of Crashing:** Determine the additional cost required to reduce the duration of each critical activity.
3. **Select Activities to Crash:** Choose the activity that offers the highest time reduction with the lowest additional cost.
4. **Update the Network Diagram:** After crashing, recalculate the project timeline and critical path. If the critical path changes, the process is repeated.
5. **Continue Crashing:** Repeat the process until the desired project duration is achieved or no further crashing is cost-effective.

Applications of Crashing:

1. When a project is running behind schedule.
2. To meet tight deadlines or compress project duration.
3. To accommodate changes in project scope or priorities.
4. When penalty costs for delays are high.

Updating a Project Network

Updating in project management refers to the continuous revision of the project's network diagram, schedule, and progress reports as the project progresses. It involves reflecting actual completion times, delays, resource adjustments, and changes in scope or schedule. **Key Aspects:**

1. **Recording Progress:** As activities are completed, their actual start and finish times are recorded, and any changes are reflected in the network diagram.
2. **Revising Estimates:** Based on the progress, the project manager may revise the estimates of remaining activity durations.
3. **Recalculating the Critical Path:** With updated time estimates, the critical path may change, and a new critical path is recalculated.

4. **Reallocating Resources:** Updating helps identify under- or over-utilized resources, leading to better resource management and adjustment.

Applications of Updating:

1. To maintain accurate and up-to-date project status.
2. For tracking progress and ensuring the project remains on schedule.
3. To assess the impact of delays and make necessary adjustments.
4. For communicating with stakeholders about the current project status.

Applications of Crashing and Updating in Project Management:

1. **Project Acceleration:** Crashing helps fast-track a project that needs to be completed earlier than originally planned. It is commonly applied in industries like construction, IT, and manufacturing, where delays could lead to penalties or financial losses.
2. **Managing Delays:** If a project is running behind schedule, crashing certain tasks and updating the network ensures that the project can still meet deadlines.
3. **Handling Changes in Project Scope:** During a project, if additional tasks or changes are introduced, crashing, and updating allow the project to be re-optimized to accommodate new requirements.
4. **Resource Optimization:** Crashing can reveal the most cost-efficient way to allocate resources, while updating helps ensure resources are utilized effectively throughout the project lifecycle.
5. **Risk Management:** Regular updating helps in risk identification by showing potential delays early. Combined with crashing, it allows for timely mitigation measures.
6. **Stakeholder Communication:** Updating provides an accurate picture of the project's status, helping project managers communicate progress, challenges, and the expected completion date to stakeholders.

Activity	Predecessor	Time estimates			Expected time $te = (o + 4m + p) \div 6$
		Opt. (o)	Normal (m)	Pess. (p)	
A	—	2	4	6	4.00
B	—	3	5	9	5.33
C	A	4	5	7	5.17
D	A	4	6	10	6.33
E	B, C	4	5	7	5.17
F	D	3	4	8	4.50
G	E	3	5	8	5.17